AI-Guided Big Data Migration: Hadoop to MRS



HUAWEI

Contents

1. **Disclaimer**
2. **Your Migration Scenario**
3. **AI-Guided Process**
4. **Expected Results**
5. **Why AI-Assisted**

AI-Assisted Migration: Usage Guidelines

Practical guidance, not a formal delivery commitment

How to work with AI for VM-to-ECS migration

How to Use This Guide

This guide shows what to ask your AI assistant, what knowledge it needs, and how to verify its work to prevent errors. AI accelerates migration planning and execution -- but does not replace expert review or customer decisions.

1. What AI Does

Planning / Analysis / Automation

- AI plans, analyzes, generates scripts, and assesses risks
- It does not replace design, expert review, testing, or decisions
- Review all outputs before acting on them

2. What to Provide

Anonymized Data Only

- Use simulated, test, or anonymized data only
- Never enter real credentials, passwords, keys, or customer data
- Respect security, compliance, and confidentiality

3. How to Work with AI

Review / Authorize / Verify

- Review every AI output before acting
- Authorize each migration step explicitly
- Keep a human in the loop for all production changes

4. When to Be Cautious

Sensitive Systems / Manual Review

- For sensitive systems (financial, defense, classified), consider manual migration
- Use strict AI oversight for compliance-heavy workloads
- Always prioritize security and sovereignty requirements

AI-assisted migration improves efficiency; it does not replace human judgment. Verify every step against real data.

Your Big Data Migration Scenario

Your Starting Point

You run a Hadoop platform on-premises or on AWS EMR, Azure HDInsight, or Google Dataproc. Moving it by hand means re-creating every component and copying petabytes without losing a table.



Your Goal

Land the whole system on MRS -- the same components (HDFS, Hive, Spark, HBase) plus all internal data -- and cut over with jobs running unchanged and no data loss.



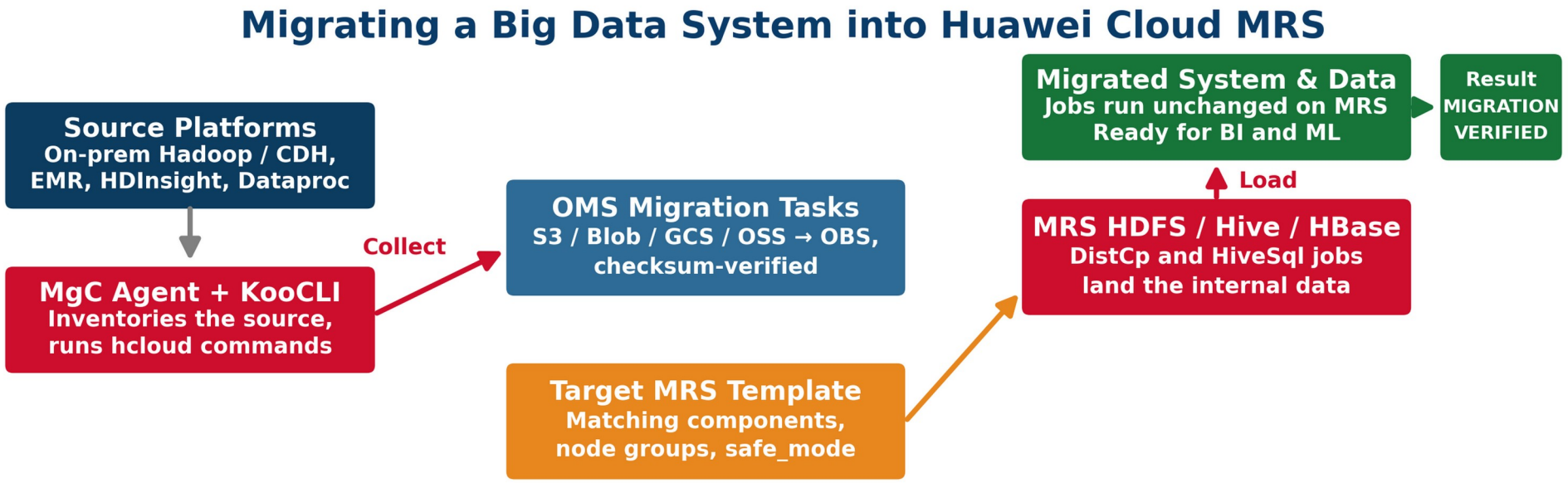
Why AI-Assisted

AI can map your source inventory to an MRS component set, generate the KooCLI and OMS commands that move the data, and reconcile every table afterwards. What used to take weeks of manual porting now takes days.



HUAWEI

Step 1: Ask AI to Inventory Your Source Cluster



Example prompt: "Inventory my source cluster -- components, versions, and data volumes -- and map it to an MRS CUSTOM cluster"
AI Should Know: How to collect a component and dataset inventory from EMR, HDInsight, Dataproc, or on-prem CDH, and map each one to MRS
Verify: Compare the AI's inventory against the source cluster. Every component your jobs use (Hive, Spark, HBase, Flink) must have a target
Prevent Hallucinations: The AI must read the real source inventory and the components your target MRS version supports, not assume them

Source: Huawei Cloud MRS and MgC Documentation (https://support.huaweicloud.com/mrs/) — Big Data Migration Guide. Accessed: Aug 2026.

HUAWEI

Step 2: Ask AI to Prepare the Target and Migration Plan

What to Ask AI

Example prompt: "Prepare an MRS CUSTOM cluster template matching my source components, plus the OMS tasks and DistCp jobs that will move the data."



AI Should Know & Verify

Which components the target MRS version supports, how OMS reaches AWS, Azure, and Google source buckets, and how DistCp loads that data into HDFS without exposing credentials.
Verify: Every source component has a target, and every dataset has a route. AI should use real flavors and real component names, not guesses.



HUAWEI

Step 3: Ask AI to Execute the Migration

MRS Migration: Two-Phase Process

Phase 1: Land & Replicate

- KooCLI creates the target MRS CUSTOM cluster
- Non-disruptive — the source cluster keeps running
- OMS moves S3, Blob, or GCS objects into OBS
- DistCp jobs load OBS data into MRS HDFS
- Monitor: task progress, job state, bytes copied

Staged →

Phase 2: Reconcile & Cut Over

- Hive metadata and HBase tables rebuilt on MRS
- MgC verifies row counts and checksums per table
- Incremental sync catches deltas since the full copy
- Pipelines re-pointed to the MRS endpoints
- Source stays available until the checks pass



On failure: keep the source running → fix the cause → re-run only the failed wave

Source: Huawei Cloud MRS and OMS Documentation (https://support.huaweicloud.com/mrs/) — Migration Execution Guide. Accessed: Aug 2026.

HUAWEI

Step 4: Ask AI to Move Data and Handle Errors

What to Ask AI

Example prompt: "Run the bulk copy into OBS, load it into HDFS with DistCp, then sync the deltas. Handle any failed task or job automatically."



AI Should Know

Common transfer failures and how to fix them. When a dataset won't copy one way, try an alternative route. How to pause and resume a migration task if needed.



Verify & Prevent Hallucinations

When the AI says a transfer is 'fixed', verify it yourself. The AI should explain which objects failed and what it did, not just say 'resolved'. Never assume a retry copied everything.



HUAWEI

Step 5: Ask AI to Validate Every Migrated Dataset

What to Ask AI

Example prompt: "Check the migration -- do row counts and checksums match the source for every table? Do our jobs produce identical output? Give me a report per dataset."



AI Should Know

What a complete migration looks like, how to compare table row counts and checksums against the source, and which jobs prove the platform works.



Verify & Prevent Hallucinations

Don't accept 'everything migrated'. Ask for proof: per-table row counts, checksum reports, job output diffs. Compare results against the source cluster. A human must approve before cutover.



HUAWEI

How to Make Your AI a Migration Expert

Teach It the Process

Tell your AI what success looks like: which datasets must match exactly, the acceptable cutover window, and the validation criteria. Clear instructions produce better results.



Give It Real Data

Connect your AI to your source inventory and your Huawei Cloud account. It should know your actual tables, buckets, and component versions -- not guess or assume.



Set Clear Expectations

Install migration skills from our GitHub repository. Point your AI to Huawei Cloud MRS, OMS, and KooCLI documentation. Works with OpenCode, Hermes, or Claude Code.



HUAWEI

Why AI-Assisted Migration

Speed & Efficiency

Porting that used to take weeks now takes days. AI maps components and generates the transfer commands in minutes. Migration planning accelerated with automated orchestration.



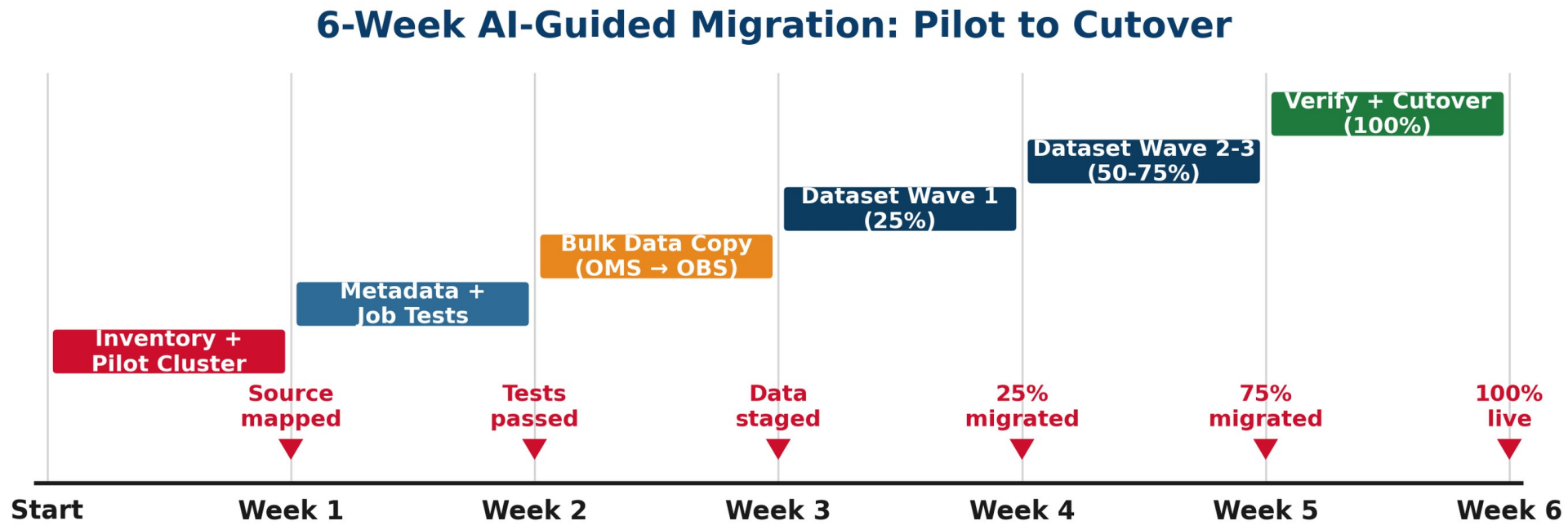
Risk Reduction

AI spots missing components and unmapped datasets before cutover. If a dataset won't copy one way, it automatically tries another. The source stays live until checks pass.



HUAWEI

Your 6-Week AI-Guided Migration Rollout



Week 1 -- Example prompt: "Inventory the source cluster and stand up a pilot MRS cluster with the matching components. Confirm everything works."
Week 2 -- Example prompt: "Migrate the Hive metadata and run our critical Hive and Spark jobs end-to-end on the pilot. Fix any issues."
Week 3-5 -- Example prompt: "Migrate datasets in waves -- 25% at a time, by criticality. Verify row counts on each wave before proceeding."
Week 6 -- Example prompt: "Sync the final deltas, run the checks, point all pipelines to MRS, and keep the source on standby."

Source: Huawei Cloud MRS and MgC Best Practices (https://support.huaweicloud.com/mrs/) — Migration Rollout Guide. Accessed: Aug 2026.

HUAWEI

Equip Your AI Assistant

Install Migration Skills:
https://github.com/binrogithub/1-3-Cloud-Adoption-Skills/tree/main/AI-Assist-Migration/MRS-KooCLI-Migration

AI Tools: OpenCode, Hermes, or Claude Code

Getting Started:
1. Install skills in ~/.opencode/skills/
2. Open your AI coding tool
3. Example prompt: "Help me migrate my Hadoop cluster to Huawei Cloud MRS using KooCLI"

Skills include: step-by-step guides, automation scripts, and runbooks

HUAWEI

Thank you.

Bring digital to every person, home and organization for a fully connected, intelligent world.

Copyright©2018 Huawei Technologies Co., Ltd.
All Rights Reserved.

The information in this document may contain predictive statements including, without limitation, statements regarding the future financial and operating results, future product portfolio, new technology, etc. There are a number of factors that could cause actual results and developments to differ materially from those expressed or implied in the predictive statements. Therefore, such information is provided for reference purpose only and constitutes neither an offer nor an acceptance. Huawei may change the information at any time without notice.

